Grade 12
June Exam Paper 1

Question 1

1.1.1. $(x+3)(x-2) = 6$

$x^2 + x - 6 = 6$

$x^2 + x - 12 = 0$ ✓A standard form

$(x+4)(x-3) = 0$

$x = -4$ ✓CA or $x = 3$ ✓CA (3)

1.1.2. $2x^2 + x - 5 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(1) \pm \sqrt{(1)^2 - 4(2)(-5)}}{2(2)}$$ ✓ correct substitution

$x \approx 1{,}35$ ✓ or $x \approx -1{,}85$ ✓ (3)

1.1.3. $\sqrt{2-x} - 4 = x$

$\sqrt{2-x} = x + 4$

$(\sqrt{2-x})^2 = (x+4)^2$ ✓m squaring both sides

$2 - x = x^2 + 8x + 16$

$x^2 + 9x + 14 = 0$ ✓A

$(x+7)(x+2) = 0$

$x = -7$ ✓CA both or $\underline{x = -2}$

N/A
✓reject (4)

1.1.4. $3^{x} - 3^{1-x} = 2$

$3^{x} - \dfrac{3}{3^{x}} = 2$ ✓

$3^{2x} - 3 = 2 \cdot 3^{x}$

$3^{2x} - 2 \cdot 3^{x} - 3 = 0$ ✓A

$(3^{x} - 3)(3^{x} + 1) = 0$

$3^{x} = 3$ or $3^{x} = -1$ ✓CA both

$\therefore x = 1$ ✓ No Solution ✓

(5)

1.1.5. $x^2 + 6x \geq 7$

$x^2 + 6x - 7 \geq 0$ ✓A

$(x+7)(x-1) \geq 0$

CV : $x = -7$; $x = 1$ ✓critical values

$\therefore x \leq -7$ ✓ or $x \geq 1$ ✓

(4)

1.2. $x - 2y = 1$ $2x^2 - xy - 5y - 3y^2 - 2 = 0 \ldots$ ②

$x = 2y + 1 \ldots$ ① ✓A

Subs ① into ②

$2(2y+1)^2 - (2y+1)y - 5y - 3y^2 - 2 = 0$ ✓m

$2(4y^2 + 4y + 1) - 2y^2 - y - 5y - 3y^2 - 2 = 0$

$8y^2 + 8y + 2 - 5y^2 - 6y - 2 = 0$

$3y^2 + 2y = 0$ ✓A

$y(3y + 2) = 0$

$y = 0$ or ✓CA both y $y = -\dfrac{2}{3}$

$\therefore x = 1$

✓CA both x $\therefore x = -\dfrac{1}{3}$

(5)

1.3. $(x-2)(x+5)=0$

$x^2+3x-10=0$ ✓ Quad

$3x^2+9x=30$ ✓ ×3.

$\therefore a=3 \qquad b=9$

$\therefore a+b=12$ ✓CA

Alternative:

$a(2)^2+b(2)=30 \qquad a(-5)^2+b(-5)=30$

$4a+2b=30 \qquad 25a-5b=30$

$2a+b=15 \ldots (1)$ ✓m $5a-b=6 \ldots (2)$

$(1)+(2)$

$7a=21$

$a=3$ ✓ a and b

$\therefore b=9$

$\therefore a+b=12$ ✓CA (3)

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Question 2

2.1. $72 \; ; \; 100 \; ; \; 120 \; ; \; 132 \; ; \ldots$

First differences: $28 \quad 20 \quad 12$

Second differences: $-8 \quad -8$ ✓m

$2a = -8$

$a = -4$

$3a + b = 28$

$3(-4) + b = 28$

$b = 40$

$a + b + c = 72$

$-4 + 40 + c = 72$

$c = 36$

$\therefore T_n = -4n^2 + 40n + 36$ ✓ ✓CA ✓CA (4)

2.2. $T_n = -4n^2 + 40n + 36$

$n_{max} = \dfrac{-40}{2(-4)} = 5$ ✓m

$T_5 = -4(5)^2 + 40(5) + 36$ ✓m

$= 136$ ✓CA

$\therefore$ max value $= 136$.

Alternate:

$T_n = -4(n^2 - 10n - 9)$ ✓

$= -4\left(n^2 - 10n + (-5)^2 - 9 - (-5)^2\right)$

$= -4\left[(n-5)^2 - 34\right]$

$= -4(n-5)^2 + 136$ ✓completed the square

$\therefore$ max value is 136 ✓CA when $n = 5$ (3)

2.3. First diff : 28 ; 20 ; 12 ; . . .

$T_n = -8n + 36$ ✓ T_n 1^{st} Diff

$T_{12} = -8(12) + 36$

$= -60$ CA ✓ T_{12}

$-60 = -4n^2 + 40n + 36$ CA ✓ m

$4n^2 - 40n - 96 = 0$

$n^2 - 10n - 24 = 0$

$(n-12)(n+2) = 0$

$n = 12$ ✓ or $n = -2$ ✓

N/A

The 12^{th} term (5)

[12]

Question 3

3.1. Geometric: ✓

$ar = 6 \ldots (1)$ $\qquad ar^4 = 162 \ldots (2)$

$(2) \div (1)$

$r^3 = 27$ ✓

$r = 3$

$\therefore a = 2$ ✓CA

Arithmetic:

$a = 2$

$d = 3$

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$

$$S_{100} = \frac{100}{2}\left[2(2) + (100-1)3\right]$$ ✓subs into correct formula

$$= 15050$$ ✓CA

(5)

3.2.1. $\sum_{K=1}^{\infty} 5\left(\frac{2-x}{3}\right)^{K-1}$ converges when $-1 < r < 1$

$$-1 < \frac{2-x}{3} < 1$$ ✓r ✓ $-1 < r < 1$

$-3 < 2 - x < 3$

$-5 < -x < 1$

$5 > x > -1$

$\therefore -1 < x < 5$ ✓

(3)

3.2.2. $5 + \frac{10}{3} + \frac{20}{9} + \cdots$

$a = 5 \qquad r = \frac{2}{3}$

$$S_n = \frac{a(1-r^n)}{1-r}$$

✓subs into correct formula

$$\frac{5\left(1-\left(\frac{2}{3}\right)^n\right)}{1-\frac{2}{3}} = 14,99$$ ✓ $S_n = 14,99$

$$1 - \left(\frac{2}{3}\right)^n = \frac{1499}{1500}$$

$$\left(\frac{2}{3}\right)^n = \frac{1}{1500}$$

$$n = \log_{\frac{2}{3}} \frac{1}{1500}$$ ✓log

$n = 18,0366\ldots$ ✓CA

$\therefore \ m = 19$ ✓CA (5)

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Question 4

4.1.1. $y = \dfrac{2}{0+1} - 3 = -1$ ✓m

$(0; -1)$ ✓ -1 (2)

4.1.2. $0 = \dfrac{2}{x+1} - 3$ ✓m

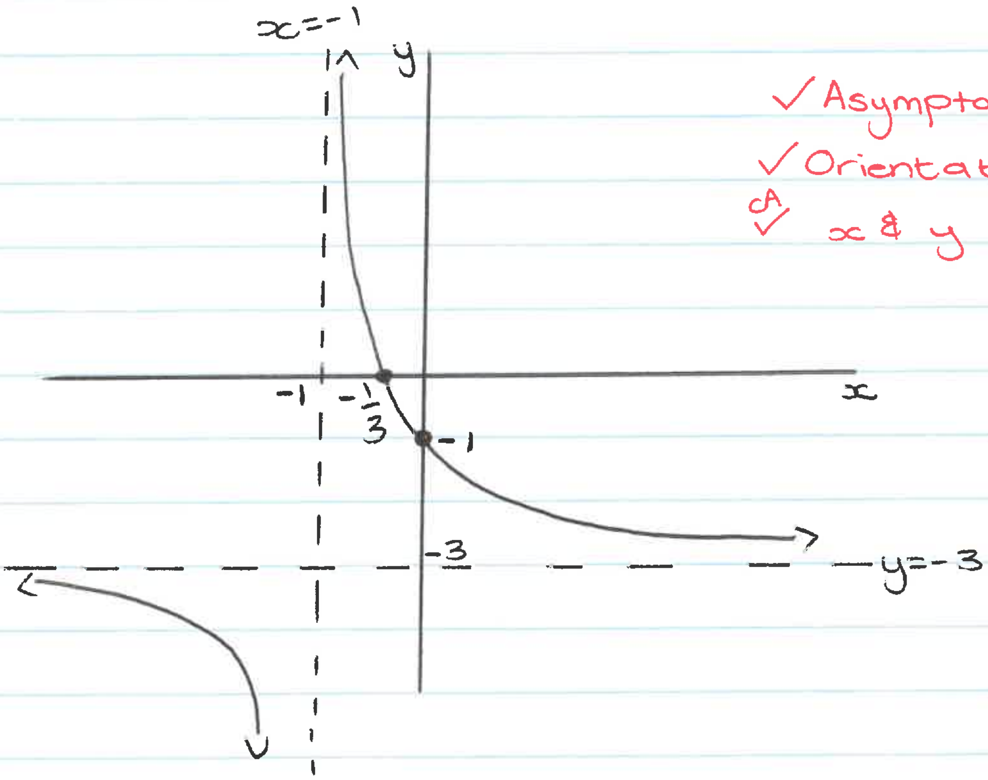
$0 = 2 - 3x - 3$

$3x = -1$

$x = -\dfrac{1}{3}$

$\left(-\dfrac{1}{3}; 0\right)$ ✓ $-\dfrac{1}{3}$ (2)

4.1.3.



✓ Asymptotes
✓ Orientation + shape
✓ CA x & y -int (3)

4.1.4 $y = -x - 4$ ✓ ✓ (2)

4.2.1. $g(x) = ab^x - 3$ ✓ $q = -3$

$-2 = ab^0 - 3$

$a = 1$ ✓M subs points

$\therefore -1 = 1 \cdot b^1 - 3$

$b = 2$

$\therefore g(x) = 2^x - 3$ ✓ $a = 1$ ✓ $b = 2$ (4)

4.2.2. $h(x) = 2 \cdot 2^x + 1$

$= 2^{x+1} + 1$

1 ← 4 ↑

$\therefore$ Translation/shift ✓ 1 unit left and ✓ 4 units up

(2)

[15]

Question 5

5.1.1. $f(x) = 2x^2 - 6x - 20$

$$x = \frac{-(-6)}{2(2)} = \frac{3}{2}$$ ✓m

$$\therefore f\left(\frac{3}{2}\right) = -\frac{49}{2}$$

$$D\left(\frac{3}{2}; -\frac{49}{2}\right)$$ ✓ ✓CA

Alternate :

$$y = 2\left(x^2 - 3x - 10 + \left(-\frac{3}{2}\right)^2 - \left(-\frac{3}{2}\right)^2\right)$$

$$= 2\left[\left(x - \frac{3}{2}\right)^2 - \frac{49}{4}\right]$$

$$= 2\left(x - \frac{3}{2}\right)^2 - \frac{49}{2}$$ ✓m completing square

$$\therefore D\left(\frac{3}{2}; -\frac{49}{2}\right)$$ ✓ ✓

(3)

5.1.2. $2x^2 - 6x - 20 = 0$ ✓m

$x^2 - 3x - 10 = 0$

$(x-5)(x+2) = 0$

$\therefore\ x = 5$ or $x = -2$ ✓A both

$A(-2;0)$ $B(5;0)$

$\therefore\ AB = 5-(-2)$

$= 7$ units ✓CA

(3)

5.1.3. $A(-2;0)$ $(-1;-12)$ ✓A $f(-1) = -12$

Average Gradient $= \dfrac{0+12}{-2+1}$ ✓m

$= -12$ ✓CA

(3)

5.1.4. $2x^2 - 6x - 20 = -p - 20$

$-p - 20 < -\dfrac{49}{2}$

$-p < -\dfrac{9}{2}$

$p > \dfrac{9}{2}$ ✓ ✓ or $4\frac{1}{2}$

Alternate:

$\Delta = b^2 - 4ac$

$= 36 - 8p$

$36 - 8p < 0$

$-8p < -36$

$p > \dfrac{9}{2}$ ✓ ✓ or $4\frac{1}{2}$

(2)

5.1.5. $x < 5 \quad ; x \neq -2$ ✓ ✓ (2)

5.1.6. $f'(x) = 4x - 6$ ✓ f'

$4x - 6 = -2$ ✓ $f' = -2$

$4x = 4$

$x = 1$

$f(1) = -24$

$P(1 ; -24)$ ✓ ✓CA (4)

5.2.1. $f(x) = \log_a x$

$-1 = \log_a \frac{1}{3}$ ✓subs S

$a^{-1} = \frac{1}{3}$ ✓

$a = 3$ (2)

5.2.2. $h: \quad x = \log_3 y$ ✓swap x & y

$y = 3^x$ ✓ (2)

5.2.3. $g(x) = -\log_3 x$ ✓ $-f(x)$ (1)

5.2.4 $0 < x \leq \frac{1}{3}$ ✓ ✓ (2)

[24]

Question 6

6.1. $A = P(1 - i)^n$

$10\,000 = 40000(1 - 15\%)^n$ ✓subs into correct formula

$\left(\frac{17}{20}\right)^n = \frac{1}{4}$

$n = \log_{\frac{17}{20}}\left(\frac{1}{4}\right)$ ✓log

$n \approx 8{,}53$ ✓ (3)

6.2. Annuity after 10 years:

$F = \frac{x\left[(1+i)^n - 1\right]}{i}$

$F = \frac{300\left[\left(1 + \frac{8{,}5\%}{12}\right)^{120} - 1\right]}{\frac{8{,}5\%}{12}}$ ✓subs correct formula

✓$n=120$

$F = 56441{,}52\ldots$ ✓

Next 10 years:

$A = P(1+i)^n$

$= 56441{,}52\ldots\left(1 + \frac{8{,}5\%}{12}\right)^{120}$ ✓subs correct formula

$= 131658{,}16$ ✓ (5)

* Accept R131658,15 if rounded off during calculations *

6.3.1. $P = \dfrac{x\left[1-(1+i)^{-n}\right]}{i}$ ✓Correct formula

$$150\ 000 = \dfrac{x\left[1-\left(1+\dfrac{15,25\%}{4}\right)^{-8}\right]}{\dfrac{15,25\%}{4}}$$

✓ A & i

✓n = 8

$x = 22107,00$ ✓

(4)

6.3.2. $OB = P(1+i)^m - \dfrac{x\left[(1+i)^m - 1\right]}{i}$ ✓m

$$= 150000\left(1+\dfrac{15,25\%}{4}\right)^4 - \dfrac{22107\left[\left(1+\dfrac{15,25\%}{4}\right)^4 - 1\right]}{\dfrac{15,25\%}{4}}$$ ✓ ✓

$= 80\ 602,00$ ✓

(4)

[16]

Question 7

7.1. $g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$

$g(x+h) = (x+h)^2 - 2(x+h) - 3$
$= x^2 + 2xh + h^2 - 2x - 2h - 3$

✓ $g(x+h)$

$g'(x) = \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - 3 - (x^2 - 2x - 3)}{h}$

✓formula

$= \lim_{h \to 0} \frac{2xh + h^2 - 2h}{h}$

$= \lim_{h \to 0} \frac{h(2x + h - 2)}{h}$ ✓ HCF

$= \lim_{h \to 0} 2x + h - 2$

$= 2x + (0) - 2$ ✓Applying limit

$= 2x - 2$ ✓

(5)

7.2.1. $y = \left(x + \frac{3}{x}\right)\left(x - \frac{3}{x}\right)$

$y = x^2 - \frac{9}{x^2}$

Binomial

$y = x^2 - 9x^{-2}$ ✓

CA ✓ CA ✓

$\therefore \frac{dy}{dx} = 2x + 18x^{-3}$ (3)

7.2.2. $y = \frac{x^2 - \sqrt{x^5}}{2x}$

$y = \frac{x}{2} - \frac{x^{\frac{3}{2}}}{2}$ ✓A

CA ✓ CA ✓

$\frac{dy}{dx} = \frac{1}{2} - \frac{3}{4}x^{\frac{1}{2}}$ (3)

$\boxed{11}$

Question 8

8.1. $y = a(x - x_1)^2 (x - x_2)$

$y = -2(x - x_A)^2 (x - 3)$ ✓

Use $P(-3; 12)$

$12 = -2(-3 - x_A)^2 (-3 - 3)$ ✓m

$1 = 9 + 6x_A + x_A^2$

$x_A^2 + 6x_A + 8 = 0$

$(x_A + 4)(x_A + 2) = 0$

$\therefore x_A = -4$ or $x_A = -2$ ca ✓both

N/A

$\therefore A(-2; 0)$ ✓

(4)

8.2. $f(x) = -2x^3 - 2x^2 + 16x + 24$

$f'(x) = -6x^2 - 4x + 16$ ✓ f'

$-6x^2 - 4x + 16 = 0$ ✓ $f' = 0$

$3x^2 + 2x - 8 = 0$

$(3x - 4)(x + 2) = 0$

$x = \frac{4}{3}$ or $x = -2$

N/A

$f\left(\frac{4}{3}\right) = \frac{1000}{27}$

$B\left(\frac{4}{3} ; \frac{1000}{27}\right)$ ✓ ✓

$\therefore BD = \frac{1000}{27}$

✓CA from y_B

$\approx 37{,}04$ units (5)

8.3. $f'(-3) = -6(-3)^2 - 4(-3) + 16$

$= -26$ ✓ $f'(-3) = -26$

$\therefore y - 12 = -26(x + 3)$ ✓m

$y - 12 = -26x - 78$

$y = -26x - 66$ ✓CA (3)

8.4. $0 < K < \frac{1000}{27}$ ✓ ✓CA from y_B (2)

8.5. $\frac{-2+\frac{4}{3}}{2} = -\frac{1}{3}$ ✓✓

Alt: $f''(x) = -12x - 4$

$-12x - 4 = 0$ ✓$f'' = 0$

$-12x = 4$

$x = -\frac{1}{3}$ ✓ (2)

8.6.1 $-2 \leq x \leq \frac{4}{3}$ CA x_B ✓ Accept $-2 < x < \frac{4}{3}$ (1)

8.6.2 $x > -\frac{1}{3}$ CA 8.5. ✓ (1)

(1)

8.7 $t = -24$ ✓A

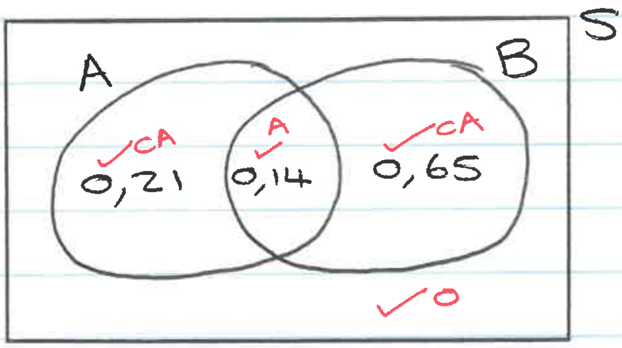
[19]

Question 9

9.1. $P(A \text{ or } B) = 1$ ✓ (1)

9.2. $P(A \text{ and } B) = 0,35 + 0,79 - 1$

$= 0,14$



(4)

9.3. $P(A) = \dfrac{n(A)}{n(S)}$

$0,35 = \dfrac{n(A)}{60}$ ✓

$\therefore n(A) = 21$ ✓ (2)

[7]

Question 10

10.1. $11!$

$= 39916800$ ✓A (1)

10.2. $\underbrace{7! \times 4!}_{✓} \times 2!$

$= 241920$ ✓×2 (2)

10.3. $\underline{7}$ $\underline{\quad 9! \quad}$ $\underline{4}$

$$\frac{7 \times 9! \times 4}{11!}$$ ✓A ✓A

$= \dfrac{14}{55}$ ✓CA (3)

[6]